

- VALERIA MIGUELEZ, PAULA MENCHÓN, *Modal expansion of Kleene algebras via twist structures.*

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Cignoli [1] showed that Kalman’s construction yields a functor from the category of bounded distributive lattices to the category of Kleene algebras, with a left adjoint that becomes an equivalence when restricted to centered Kleene algebras satisfying the interpolation property.

Jalali [2] extended this representation to all Kleene algebras by means of *Kleene triples* $\langle L, R, F \rangle$, where L is a distributive lattice, R is an independence relation given as the complement of contact relations, and F is a Boolean filter of L .

Motivated by this framework and by recent work on modal Nelson algebras [3], we introduce modal operators on the distributive lattice L that interact in a precise way with the contact relation and the Boolean filter of Kleene triples. This allows us to represent modal Kleene algebras through twist structures. Moreover, we establish a Priestley-type topological duality for these modal expansions. We also discuss applications to Kleene algebras with implication and the particular case of Nelson algebras.

[1] R. CIGNOLI, *The class of Kleene algebras satisfying an interpolation property and Nelson algebras*, **Algebra Universalis**, vol. 23 (1986), pp. 262–292.

[2] A. A. JALALI, *A representation of Kleene algebras and the characterisation of projective Kleene and De Morgan algebras*, **Rendiconti del Circolo Matematico di Palermo. Serie II. Supplemento**, no. 29 (1992), pp. 435–474.

[3] P. MENCHÓN and R. O. RODRIGUEZ, *Twist-structures isomorphic to modal Nelson lattices*, **arXiv preprint**, arXiv:2504.08109 (2025).